ISSN 0005-1179, Automation and Remote Control, 2020, Vol. 81, No. 8, pp. 1499–1506. © Pleiades Publishing, Ltd., 2020.
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TOPICAL ISSUE

Phase Control for the Dynamics of Connected Rotators

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Received July 23, 2019
Revised October 18, 2019
Accepted January 30, 2020

Abstract—We study the dynamics of rotational motions in a system of two asymmetrically coupled pendulum-type systems, investigating the mechanisms of losing stability for in-phase rotational motion. We analyze the scenario where chaotic dynamics appears depending on the values of the control parameters.

Keywords: rotator, phase control, synchronization, rotational mode, chaos

DOI: 10.1134/S0005117920080111

1. INTRODUCTION

The study of collective behavior in networks of connected elements is an attractive and important area of nonlinear dynamics, relevant from the point of view of both theory and applications [1–4]. It is known that even with a weak coupling, elements of ensembles can tend to achieve a common rhythm of operation, i.e., to synchronize [1]. A rather wide class of objects considered in physics, radio engineering, electronics, and other fields of natural science can be described using models of coupled pendulum systems [5]. Despite the simplicity of these models, they are used not only to describe mechanical objects [6] but also for various processes in molecular biology [7–9], semiconductor structures [10], and more. This model can also be considered as a basic model in theoretical studies of coupled Josephson contacts [11–13], as well as phase synchronization systems [3, 4, 14, 15].

2. MODEL DESCRIPTION

In this work, we consider the behavior of an ensemble of two partial phase synchronization systems connected in parallel through phase mismatch signals [3, 4]. A block diagram of this ensemble is shown in Fig. 1. The mathematical model of a system of two such objects can be represented as a system of pendulum-type equations:

$$\begin{aligned} \ddot{\varphi}_1 + \lambda\dot{\varphi}_1 + \sin\varphi_1 &= \gamma + \kappa_1 \sin(\varphi_2 - \varphi_1), \\ \ddot{\varphi}_2 + \lambda\dot{\varphi}_2 + \sin\varphi_2 &= \gamma + \kappa_2 \sin(\varphi_1 - \varphi_2). \end{aligned} \tag{1}$$

Here λ is the signal attenuation coefficient, γ is the ratio of the initial to maximum frequency mismatch, κ_1, κ_2 are the signal gain parameters characterizing the strength of coupling between systems.

An ensemble of two symmetrically connected identical pendulums was considered in [16]. The dynamics of non-identical pendulums (with different values of γ) were studied in [17]. Note that system (1) can be used to describe the behavior of an ensemble of globally coupled rotators where

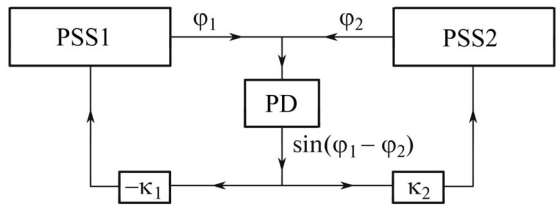

Fig. 1. Block diagram of a pair of phase synchronization systems (PSS1 and PSS2) connected in parallel via phase mismatch signals (PD—phase detector).

two clusters are formed with different numbers of mutually synchronous elements (N_1 and N_2) [18]. Since N_1 and N_2 are different, the coupling between the clusters is naturally asymmetric

We study the dependency of the behavior of system (1) on the degree of asymmetry in the coupling. To do this, consider system (1) in the form

$$\begin{aligned} \ddot{\varphi}_1 + \lambda\dot{\varphi}_1 + \sin\varphi_1 &= \gamma + K\sin(\varphi_2 - \varphi_1), \\ \ddot{\varphi}_2 + \lambda\dot{\varphi}_2 + \sin\varphi_2 &= \gamma + \beta K\sin(\varphi_1 - \varphi_2), \end{aligned} \tag{2}$$

where now K is the coupling parameter and β is the parameter characterizing the degree of asymmetry of the coupling. This type of communication is an additional way of controlling dynamic modes, including synchronous modes in various kinds of technical devices, in particular in phase synchronization systems [3, 4]. As we will show below, with a certain choice of control parameters such a system can have both in-phase and out-of-phase rotational modes.

3. IN-PHASE MODE AND STABILITY

System (2) has an in-phase periodic motion: the coordinates coincide with each other, i.e., $\varphi_1(t) = \varphi_2(t) = \phi_s(t)$. Moreover, $\phi_s(t)$ satisfies the equation

$$\ddot{\phi}_s + \lambda\dot{\phi}_s + \sin\phi_s = \gamma.$$

Let us determine for what values of the control parameters the in-phase rotational mode of system (2) loses stability. To do this, we linearize system (2) with respect to the in-phase rotational motion ϕ_s. We represent the phases of rotators in the form

$$\begin{aligned} \varphi_1 &= \phi_s + \delta\varphi_1, \\ \varphi_2 &= \phi_s + \delta\varphi_2, \end{aligned}$$

getting a system of equations in variations

$$\begin{aligned} \delta\ddot{\varphi}_1 + \lambda\delta\dot{\varphi}_1 + \cos(\phi_s)\delta\varphi_1 &= K(\delta\varphi_2 - \delta\varphi_1), \\ \delta\ddot{\varphi}_2 + \lambda\delta\dot{\varphi}_2 + \cos(\phi_s)\delta\varphi_2 &= \beta K(\delta\varphi_1 - \delta\varphi_2). \end{aligned}$$

We introduce a new value $\eta = \delta\varphi_2 - \delta\varphi_1$, which is the mismatch of phase increments in the vicinity of ϕ_s, and obtain a differential equation with respect to η:

$$\ddot{\eta} + \lambda\dot{\eta} + (\cos\phi_s + (1+\beta)K)\eta = 0. \tag{3}$$

A detailed analysis of Eq. (3) has been presented in [16, 19], where it was shown that there exists a range of values of the parameter K ($K_1 < K < K_2$) where the in-phase rotational motion becomes unstable. The values K_1, K_2 in this case are given by the expression

$$K_{1,2} = \frac{1}{1+\beta}\left[\frac{\gamma^2}{\lambda^2} \pm 2\sqrt{1-\gamma^2} + \frac{1}{2}\frac{\lambda^2}{\gamma^2}\right] + O\left(\frac{\lambda^4}{\gamma^4}\right). \tag{4}$$

Inside the interval $\Delta K = K_2 - K_1$ both asynchronous periodic (with different periods) and chaotic movements can occur. It follows from (4) that an increase in the asymmetry of the coupling leads to a reduction of the instability region in the in-phase mode. In this case, the instability interval ΔK shifts to the region of small coupling coefficients K.

4. FINDING REGULAR ROTATIONAL MODES

Let us describe the numerical methods used to search for regular rotational motions existing in the base model (2) and determine their linear stability. To begin with, we note that the procedure described below is actually a modification of the scheme for finding closed limit cycles in nonlinear dynamical systems and uses the methods described in [20]. We introduce the notation $\{\varphi_n(t)\}$—here and below it denotes a row vector, where $n = 1, 2$. The main idea of this method is as follows. Any representative $\{\varphi_n(t)\}$ of the desired class of trajectories is characterized primarily by its period T (which is unknown and must be found at the end of the computational procedure) and the number k that determines how many times each component from the set of cyclic coordinates $\{\varphi_n(t)\}$ will change by 2 π over the time period T. Based on this, let us consider the mapping $\{\varphi_n(0), \dot{\varphi}_n(0)\} \to \{\varphi_n(T), \dot{\varphi}_n(T)\}$ and construct the following vector:

$$p(T, \{\varphi_{0n}, \dot{\varphi}_{0n}\}) = \{\varphi_n(T, \{\varphi_{0n}, \dot{\varphi}_{0n}\}) - \varphi_{0n} - 2\pi k, \dot{\varphi}_n(T, \{\varphi_{0n}, \dot{\varphi}_{0n}\}) - \dot{\varphi}_{0n}\},$$

where $\{\varphi_n(t), \dot{\varphi}_n(t)\}$ is a solution for system (2) with initial conditions $\{\varphi_{0n}, \dot{\varphi}_{0n}\}$, i.e. $\{\varphi_n(0), \dot{\varphi}_n(0)\} = \{\varphi_{0n}, \dot{\varphi}_{0n}\}$.

Using the multidimensional function $p(T, \{\varphi_{0n}, \dot{\varphi}_{0n}\})$ defined in a similar way, we can formulate a condition that makes it possible to find $\{\varphi_n(t)\}$ and T. This condition is that all components of the vector $p(T, \{\varphi_{0n}, \dot{\varphi}_{0n}\})$ are equal to zero. As a result, we come to the conclusion that it is necessary to choose such values T and $\{\varphi_n(t)\}$ that will satisfy

$$p(T, \{\varphi_{0n}, \dot{\varphi}_{0n}\}) = 0. \tag{5}$$

In other words, we can say that the task now is to find a fixed point for the mapping $\{\varphi_n(0), \dot{\varphi}_n(0)\} \to \{\varphi_n(T), \dot{\varphi}_n(T)\}$, taking into account the cyclical nature of $\{\varphi_n(t)\}$ [20]. Due to the invariance of system (2) with respect to time shifts, one of the values $\{\varphi_{0n}\}$ can be equal to zero without loss of generality, thus equating the number of unknowns and the number of relations in (3). To find the roots of the set of Eqs. (3), it makes sense to use the Newton's method since it has high efficiency. Continuing these solutions with respect to parameter β in the instability interval of the in-phase mode, we can trace the entire family of nontrivial periodic motions and analyze their bifurcations.

To study the linear stability of arbitrary (2π-, 4π-, 8π-, and so on) periodic rotational motions (taking into account their cyclical nature) of the dynamical system (2) we introduce small perturbations $\delta\varphi_n(t)$: $\{\varphi_n(t) = \phi_n(t) + \delta\varphi_n(t)\}$, where $\phi_n(t)$ is the considered periodic motion. As a result of the linearization procedure, we obtain the following equations for perturbations $\{\delta\varphi_n(t)\}$:

$$\delta\ddot{\varphi}_1 + \lambda\delta\dot{\varphi}_1 + \cos(\phi_1)\delta\varphi_1 = K\cos(\phi_2 - \phi_1)(\delta\varphi_2 - \delta\varphi_1),$$
$$\delta\ddot{\varphi}_2 + \lambda\delta\dot{\varphi}_2 + \cos(\phi_2)\delta\varphi_2 = \beta K\cos(\phi_1 - \phi_2)(\delta\varphi_1 - \delta\varphi_2).$$

Further analysis can be carried out in the framework of Floquet theory. The stability of the considered motions is determined by the spectrum of eigenvalues of the monodromy matrix (Floquet operator) $M(T)$, which is given by the expression

$$\{\delta\varphi_n(T), \delta\dot{\varphi}_n(T)\}^T = M\{\delta\varphi_n(0), \delta\dot{\varphi}_n(0)\}^T.$$

The eigenvalues μ_m (here and below $m = \overline{1,4}$) of the matrix $M(T)$ are Floquet multipliers, which are associated with Floquet exponents q_m of the periodic solution $\{\phi_n(t)\}$ with ratios $m = \exp(iq_m)$.

Thus, to determine the stability of each considered movement, it suffices to calculate μ_m. If $|\mu_m| \leq 1$ for all m, and then the rotational mode is linearly stable. It is worth noting that one of the eigenvalues μ_m must always be strictly equal to one, since $\{\dot{\varphi}_n(t)\}$ belongs to the family of periodic motions (taking into account its cyclic nature). Therefore, there is an additional opportunity to verify that the solution $\{\phi_n(t)\}$ found using the above procedure belongs to the considered class of limit rotations. If at least one of the Floquet multipliers μ_m is located on the complex plane outside the unit circle, then the rotational mode is linearly unstable.

5. OUT-OF-PHASE REGULAR AND CHAOTIC ROTATIONAL MODES

As a result of numerical modeling in the region of parameters where there is no stability of the in-phase rotational motion (see expression (4)), we have studied steady-state rotational motions and their bifurcations.

In order to characterize the degree of deviation from the in-phase mode, we introduce the value $\xi = \max\limits_{0<t<T} |\dot{\varphi}_1(t) - \dot{\varphi}_2(t)|$, where T is the period of rotational movements. Here and below, the external influence parameter is chosen to be equal to $\gamma = 0.97$.

Figures 2a–2c and Figs. 3a and 3b show the local maxima of the instantaneous frequencies of the oscillators $\dot{\varphi}_1(t)$ and $\dot{\varphi}_2(t)$. Round markers show the values $\max(\dot{\varphi}_1(t))$, cross markers show $\max(\dot{\varphi}_2(t))$. In Figs. 2d–2f and Figs. 3c and 3d we show dependencies of parameter ξ on the param-

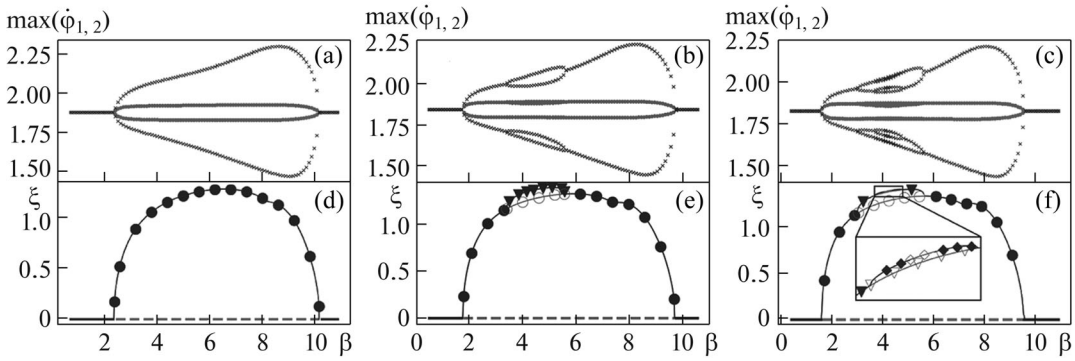

Fig. 2. (a)–(c) Local maxima $\dot{\varphi}_1, \dot{\varphi}_2$. (d)–(f) Bifurcation diagrams of rotational modes.

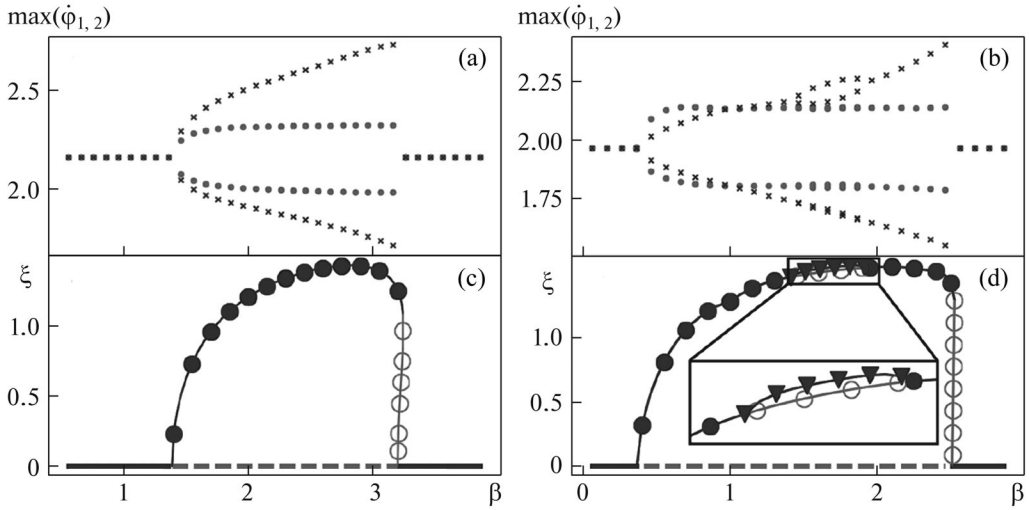

Fig. 3. (a), (b) Local maxima $\dot{\varphi}_1, \dot{\varphi}_2$. (c), (d) Bifurcation diagrams of rotational modes.

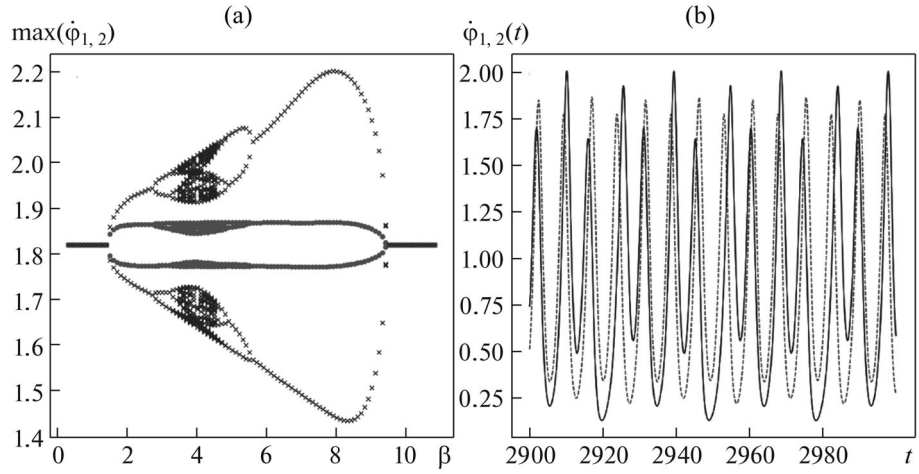

Fig. 4. (a) Local maxima $\dot{\varphi}_1, \dot{\varphi}_2$. (b) Dynamic chaos mode for $\beta = 3.977$.

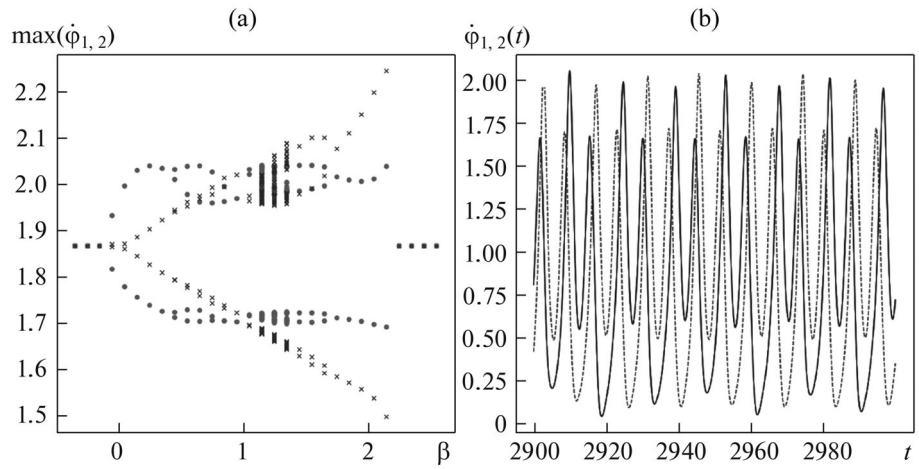

Fig. 5. (a) Local maxima $\dot{\varphi}_1, \dot{\varphi}_2$. (b) Dynamic chaos mode for $\beta = 1.25$.

eter β. Shaded markers correspond to stable rotational movements, empty markers—to unstable ones. In this case, round, triangular, and rectangular markers show 4π-, 8π-, and 16π-periodic rotational modes respectively. The line without markers corresponds to the in-phase 2π-periodic rotational mode, the solid line—to the stable mode, dotted—to the unstable mode. For $K = 0.06$, $\lambda = 0.77$ (Figs. 2a and 2d) as parameter β increases, the in-phase periodic rotational motion undergoes a period doubling bifurcation at $\beta \approx 2.3$. Moreover, the stable in-phase 2π-periodic motion gives rise to a stable 4π-periodic motion, while the 2π-periodic motion loses its stability. Then, at $\beta \approx 10.2$ as a result of the reverse doubling bifurcation the stable 4 π-periodic motion merges with the unstable in-phase 2 π-periodic motion, and the in-phase rotational motion again becomes stable. At $\lambda = 0.8$ (Figs. 2b and 2e) the now out-of-phase 4π-periodic motion undergoes a period doubling bifurcation. and at the same time an 8π-periodic rotational motion is born, and the 4π-periodic motion loses its stability. Further, for larger values of the dissipation parameter $\lambda = 0.81$ (Figs. 2c and 2f), when the parameter β increases, the system undergoes several period doubling bifurcations, as a result of which the number of unstable rotational motions increases. Figures 4 and 5 show local maxima of instantaneous frequencies $\dot{\varphi}_1(t), \dot{\varphi}_2(t)$. The round (cross-shaped) markers indicate the values $\max(\dot{\varphi}_1(t))$ ($\max(\dot{\varphi}_2(t))$) (Figs. 4a and 5a) and temporal realizations $\dot{\varphi}_1(t), \dot{\varphi}_2(t)$ (Figs. 4b and 5b). The dotted line corresponds to $\dot{\varphi}_1(t)$, solid line, to $\dot{\varphi}_2(t)$. For $\lambda = 0.816$ (Fig. 4), we see that as a result of the cascade of period doubling bifurcations there arises a range of values

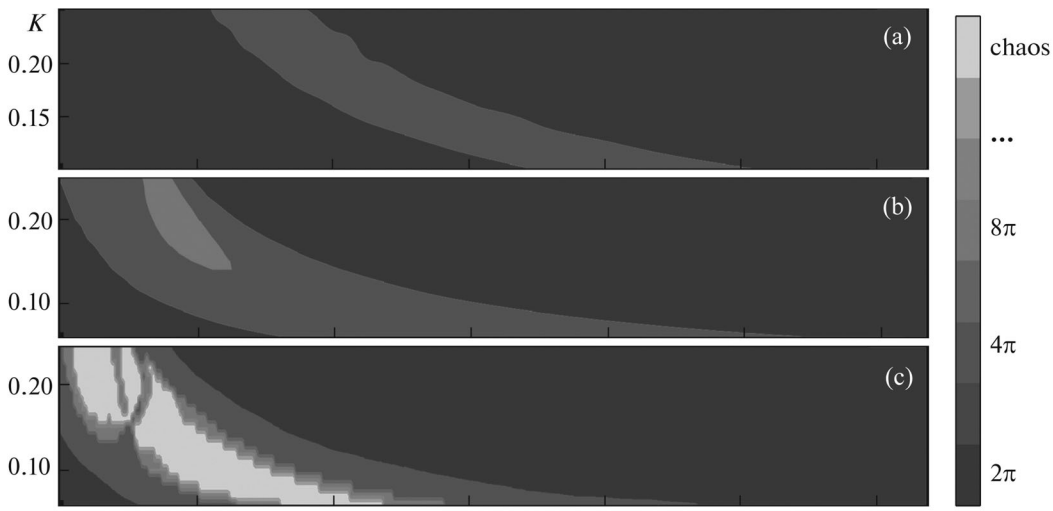

Fig. 6. Type of rotational periodic modes depending on the values of parameters K, β for $\gamma = 0.97$. (a) $\lambda = 0.67$, (b) $\lambda = 0.73$, (c) $\lambda = 0.82$.

of the parameter β ($3.53 < \beta < 4.53$) inside which dynamic chaos is observed in the system [21]. Now, consider the case where $K = 0.21$ and $\lambda = 0.6$ (Figs. 3a and 3c). As a result of the period doubling bifurcation, the in-phase rotational motion here also loses stability at $\beta \approx 1.39$. In this case, a stable 4π-periodic motion is born, the difference of which from the in-phase mode increases with increasing parameter β. However, for $\beta > 3.22$, system (2) again returns to the situation where the state $\varphi_1(t) = \varphi_2(t)$ is established. On the bifurcation diagram (Fig. 3c) we can see that such a change in the behavior of the model (2) occurs abruptly when passing through the point c $\beta = 3.22$. This hard disappearance effect is due to the existence of an unstable rotational motion in the interval $3.199 < \beta < 3.24$.

To analyze the changes that are taking place, consider the dependence of the parameter ξ on the parameter β (Figs. 3a and 3c). It can be seen that in addition to stable periodic motions, there exists also an unstable out-of-phase 4π-periodic motion which is born from the in-phase unstable 2π-periodic motion as a result of a subcritical period doubling bifurcation ($\beta \approx 3.199$), and at the same time the in-phase 2 π-periodic movement becomes stable again. Further, as the parameter β increases stable and unstable 4 π-periodic rotational movements merge and disappear as a result of a saddle-node bifurcation. With a further increase in parameter β in system (2), only in-phase rotational mode becomes possible. Thus, the system exhibits a bistability effect of rotational modes. For $\lambda = 0.71$ (Figs. 3b and 3d), the situation is similar to the previous case: there is a 4π-periodic movement and a period doubling bifurcation ($\beta \approx 1.38$) occurs, as a result of which a stable 8π-periodic movement is born, while the 4π-periodic movement loses its stability. At $\lambda = 0.779$ (Fig. 5), as a result of a cascade of period doubling bifurcations there appears a range of values of parameter β ($1.15 < \beta < 1.35$) for which the system exhibits a dynamic chaos mode.

Figure 6 shows maps of rotational modes that contain the type of steady-state rotational motion realized in the system depending on the parameters K and β. Consider the case $\lambda = 0.67$ (Fig. 6a), where in addition to the in-phase rotational motion in the considered parameter range there exists only a stable out-of-phase 4π-periodic rotational mode. With an increase of λ to the value 0.73 (Fig. 6b), apart from a stable 4π-periodic rotational mode, for some K and β the system can exhibit an 8π-periodic out-of-phase rotational mode, while the 4π-periodic rotational motion loses stability. At $\lambda = 0.82$ (Fig. 6c), a dynamic chaos regime can be observed in the system, resulting from a cascade of period doubling bifurcations.

Note that for small values of the coupling parameter K the instability of the in-phase mode occurs with a larger value of the asymmetric coupling parameter β.

6. CONCLUSION

In this work, we have considered rotational dynamics in coupled phase synchronization systems. We have shown that in a system with asymmetric coupling there exists a range of parameter values for which the in-phase rotational motion is unstable, and out-of-phase periodic and chaotic rotations are realized. We have analytically shown that an increase in the asymmetry of the coupling leads to a reduction in the instability region for the in-phase mode $\Delta K = K_2 - K_1$. In this case, the instability interval ΔK shifts to the region of small coupling coefficients K. For instability to occur at small values of the coupling parameter, a large degree of asymmetry is necessary.

The loss of stability occurs as a result of period doubling bifurcation (forward and reverse). During the reverse period doubling bifurcation, the system exhibits a rigid transition to an out-of-phase regime. In the case of large values of the attenuation parameter, chaotic rotations occur as a result of a cascade of period doubling bifurcations.

We have presented numerical methods that allow to find and determine the linear stability of regular rotational regimes.

FUNDING

This work was supported by the Russian Science Foundation, project no. 19-12-00367.

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This paper was recommended for publication by B.T. Polyak, a member of the Editorial Board